

Privacy-Preserving Natural Language Processing

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Lecture 5 – Local DP and Randomized Response

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Recap

- 1 Recap
- 2 Math refresher
- 3 Local differential privacy
- 4 Using local DP in NLP

What we covered so far

- For provably private data analysis we need randomized algorithms
- Central (with a trusted curator) pure $(\epsilon, 0)$ differential privacy
- Laplace mechanism: numeric queries, ℓ_1 sensitivity,
- Exponential mechanism: 'any-range' queries (arbitrary sets), utility function and its sensitivity

Today

- ...
- Central (with a **trusted curator**) pure $(\varepsilon, 0)$ differential privacy
- ...

What if we don't trust anyone?

Math refresher

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Bernoulli trial (Bernoulli distribution, Bernoulli R.V., etc.)

Suppose that we run an experiment that succeeds with probability θ and fails with probability $1 - \theta$

A Bernoulli (or an indicator) random variable Y has the following probability distribution on $k \in \{0, 1\}$

$$\Pr(Y = 1) = \theta$$

$$\Pr(Y = 0) = 1 - \theta$$

which can be also expressed as

$$\Pr(Y = k) = \theta^k (1 - \theta)^{1-k}$$

Local differential privacy

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Motivation

The notion of neighboring datasets, revisited

Database of size 1, replacing one person

Can we ask numeric queries on database of size 1?

Sure!

Example query: How many classes did you skip (and binge-watched Netflix instead)?

True value: between 0 and 15

You just need the ℓ_1 sensitivity for Laplace mechanism, decide on the ϵ and report

DP enables learning about population, not individuals!

What I'm really interested in is some statistics, e.g., the average number of lectures skipped across all students.

But 'privatizing' the 'raw' values shown previously removes the need for a trusted curator.

It comes with extra price (to be discussed later)

Simplifying queries

Now we know that we can privately (DP) answer statistical queries without a trusted curator

Let's simplify our example even more

Reporting single bit

Each person has a secret bit, e.g. "I took illegal drugs"

The query I might be interested in is 'What percentage of students take illegal drugs'

How to report a single bit

From true value to reported value

Two extremes

- Just report the true value
- Completely random answer (e.g., toss a coin)

Something between these two extremes

How to report a single bit: From true value to reported value

Idea: Report the true value with some probability, 'flip' and report otherwise

How to do the above 'with some probability' exactly?

Again: Use a randomized algorithm!

- Step 1: Draw a random value, either 0 or 1, from a Bernoulli distribution
- Step 2: If 1, report true value, if 0 report flipped value

(1 is usually associated with 'true' or 'success' and 0 with 'false' or 'failure')

How to parametrize this Bernoulli distribution?

Step 1: Draw 0 or 1 from $\text{Ber}(\theta)$, Step 2: If 1, report true value, if 0 report flipped value

What θ would we need for modeling the extremes? (by the way: θ is publicly known! No 'security through obscurity'!)

Extreme 1: Just report the true value

$\text{Ber}(\theta = 1)$ always true value (no privacy)

Extreme 2: Completely random answer

$\text{Ber}(\theta = 0.5)$ completely random $\rightarrow$ no matter what the raw value is, this is equivalent to reporting a random value

How to parametrize this Bernoulli distribution?

We saw that θ controls the privacy strength and want

$$0.5 < \theta < 1.0$$

where $0.5 \leftarrow \theta$ — more privacy

and $\theta \rightarrow 1.0$ — less privacy

How do we control privacy strength in DP?

$0 \leftarrow \varepsilon$ — more privacy

$\varepsilon \rightarrow \infty$ — less privacy

We need to find a function that maps ε to θ (and also satisfies DP)

First attempt

We want $0.5 < \theta < 1.0$ as a function of $0 \leq \varepsilon \leq \infty$

$\lim_{\varepsilon \rightarrow \infty} 1 = 1$ so does $\lim_{\varepsilon \rightarrow \infty} \frac{\varepsilon}{\varepsilon} = 1$ as well as $\lim_{\varepsilon \rightarrow \infty} \frac{\varepsilon}{\varepsilon+1} = 1$

But it doesn't work for $\lim_{\varepsilon \rightarrow 0}$ yet because $\lim_{\varepsilon \rightarrow 0} \frac{\varepsilon}{\varepsilon+1} \neq 0.5$

The easiest fix: Just add 1!

$\lim_{\varepsilon \rightarrow 0} \frac{\varepsilon+1}{\varepsilon+1+1} = 0.5$ and $\lim_{\varepsilon \rightarrow \infty} \frac{\varepsilon+1}{\varepsilon+1+1} = 1$

So let's try with $\theta = \frac{\varepsilon+1}{\varepsilon+2}$

Try to prove DP with our Bernoulli trial

For any 'neighboring' $D, D' \in \{0, 1\}$ and any $y \in \{0, 1\}$ the ε -DP definition must hold:

$$\frac{\Pr(\mathcal{M}(D) = y)}{\Pr(\mathcal{M}(D') = y)} \leq \exp(\varepsilon)$$

So let's try with Bernoulli trial and report the true with probability $\theta = \frac{\varepsilon+1}{\varepsilon+2}$ (and flip with probability $1 - \theta$)

Let $D = 0, D' = 1, y = 0$ (D' flipped from 1 to 0):

$$\begin{aligned}\frac{\Pr(y = 0|D = 0)}{\Pr(y = 0|D' = 1)} &= \frac{\theta}{1 - \theta} = \frac{\frac{\varepsilon+1}{\varepsilon+2}}{1 - \frac{\varepsilon+1}{\varepsilon+2}} = \frac{\frac{\varepsilon+1}{\varepsilon+2}}{\frac{\varepsilon+2-(\varepsilon+1)}{\varepsilon+2}} = \frac{\varepsilon+1}{\varepsilon+2} \cdot \frac{\varepsilon+2}{1} \\ &= \varepsilon + 1 \leq \exp(\varepsilon)\end{aligned}$$

The same holds also for $D = 1, D' = 0, y = 1$ (D' flipped).

For not flipping, the ratio is 1, so the inequality also holds!

Great! We designed a DP algorithm for reporting a single bit (locally)

But can we do any better (with better utility?)

Improving the inequality

Our first attempt

$$\frac{\Pr(y=0|D=0)}{\Pr(y=0|D'=1)} = \frac{\theta}{1-\theta} = \frac{\frac{\varepsilon+1}{\varepsilon+2}}{1-\frac{\varepsilon+1}{\varepsilon+2}} = \varepsilon + 1 \leq \exp(\varepsilon)$$

We can approximate $\exp(\varepsilon)$ with a line around 0

$$\exp(\varepsilon) \geq 1 + \varepsilon$$

but also with a polynomial of degree two

$$\exp(\varepsilon) \geq 1 + \varepsilon + \frac{\varepsilon^2}{2}$$

We get equality with the very definition of using power series $\exp(\varepsilon) := \sum_{k=0}^{\infty} \frac{\varepsilon^k}{k!}$

Improving the inequality

Our first attempt

$$\frac{\Pr(y=0|D=0)}{\Pr(y=0|D'=1)} = \frac{\theta}{1-\theta} = \frac{\frac{\varepsilon+1}{\varepsilon+2}}{1-\frac{\varepsilon+1}{\varepsilon+2}} = \varepsilon + 1 \leq \exp(\varepsilon)$$

What if we replace $\varepsilon + 1$ with $\exp(\varepsilon)$ and worked it out backwards?

Our second attempt

$$\frac{\Pr(y=0|D=0)}{\Pr(y=0|D'=1)} = \frac{\theta}{1-\theta} = \dots? \dots = \exp(\varepsilon) \leq \exp(\varepsilon)$$

Second attempt: Replace $\varepsilon + 1$ with $\exp(\varepsilon)$ backwards

First attempt recap (now rewriting with $2 = 1 + 1$)

$$\begin{aligned}\frac{\Pr(y = 0|D = 0)}{\Pr(y = 0|D' = 1)} &= \frac{\frac{\varepsilon+1}{\varepsilon+1+1}}{1 - \frac{\varepsilon+1}{\varepsilon+1+1}} = \frac{\frac{\varepsilon+1}{\varepsilon+1+1}}{\frac{\varepsilon+1+1-(\varepsilon+1)}{\varepsilon+1+1}} \\ &= \frac{\varepsilon + 1}{\varepsilon + 1 + 1} \cdot \frac{\varepsilon + 1 + 1}{1} = \varepsilon + 1 \leq \exp(\varepsilon)\end{aligned}$$

$$\begin{aligned}\frac{\Pr(y = 0|D = 0)}{\Pr(y = 0|D' = 1)} &= \frac{\frac{\exp(\varepsilon)}{\exp(\varepsilon)+1}}{1 - \frac{\exp(\varepsilon)}{\exp(\varepsilon)+1}} = \frac{\frac{\exp(\varepsilon)}{\exp(\varepsilon)+1}}{\frac{\exp(\varepsilon)+1-\exp(\varepsilon)}{\exp(\varepsilon)+1}} \\ &= \frac{\exp(\varepsilon)}{\exp(\varepsilon) + 1} \cdot \frac{\exp(\varepsilon) + 1}{1} = \exp(\varepsilon) \leq \exp(\varepsilon)\end{aligned}$$

Second attempt: Replace $\varepsilon + 1$ with $\exp(\varepsilon)$ backwards

We found the tightest θ

$$\frac{\Pr(y = 0|D = 0)}{\Pr(y = 0|D' = 1)} = \frac{\theta}{1 - \theta} = \frac{\frac{\exp(\varepsilon)}{\exp(\varepsilon)+1}}{1 - \frac{\exp(\varepsilon)}{\exp(\varepsilon)+1}} \underbrace{\leq}_{\text{(actually =)}} \exp(\varepsilon)$$

Recap our algorithm:

- Step 1: Draw a random value, either 0 or 1, from a Bernoulli distribution
- Step 2: If 1, report true value, if 0 report flipped value

We found $\theta = \frac{\exp(\varepsilon)}{\exp(\varepsilon)+1} = \frac{1}{1+\exp(-\varepsilon)}$ makes it $(\varepsilon, 0)$ -DP

We discovered the "Randomized response" algorithm

- Step 1: Draw a random value, either 0 or 1, from a Bernoulli distribution
- Step 2: If 1, report true value, if 0 report flipped value

Parametrize Bernoulli with $\theta = \frac{\exp(\varepsilon)}{\exp(\varepsilon)+1} = \frac{1}{1+\exp(-\varepsilon)}$

Homework: Prove that Randomized response is $(\varepsilon, 0)$ -DP
(just do what we did for any pair of D, D', y)

We discovered the "Randomized response" algorithm

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Local differential privacy

From single bit to bit strings and beyond

From single bit to bit strings

- RR limited to a single boolean variable
- Can we protect a bit-string?

Two views on the process

View 1: Ask 'What is your i -th bit?' for $i = 1, \dots, n$ consecutively

- Each answer protected independently with RR (ε)
- Total spent privacy budget: $n \cdot \varepsilon$

View 2: Naive composition of n $(\varepsilon, 0)$ -DP mechanisms

- ε 's simply sum up

Bad consequences for the utility

- Increasing number of revealed bits has bad consequences for utility
- To keep overall budget at ε , each bit's budget is ε/n
- Increases probability of *not* reporting the truth
- Results in much noisier output vector

Generalized randomized response

- Extend range from binary $D \in \{0, 1\}$ to finite set of k categorical values
- Set $C = \{1, 2, \dots, k\}$
- True value $t \in C$ reported with probability

$$\theta = \frac{\exp(\varepsilon)}{\exp(\varepsilon) + |C| + 1}$$

- With probability $1 - \theta$, report random value sampled uniformly from $C \setminus \{t\}$

Properties of generalized randomized response

Properties

- Generalized randomized response is $(\varepsilon, 0)$ -DP
- If $C = \{0, 1\}$, it becomes the usual binary randomized response'

Using local DP in NLP

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Motivation for Local DP in NLP

- Each data-owner privatizes their data point before releasing to public
- Possible use: Privacy protection of training data for classification tasks
- Assumption: n individuals each holding data-point (text document + categorical label)
- Example: Reviews for binary sentiment analysis (e.g., IMDB)
- Sensitive example: Drug reviews (review + stars)
- Goal: Train model to classify review into two or multiple classes

What are we protecting?

- Single document d (e.g., "I had really bad headaches ...")
- Associated label $y \in \{0, 1, 2\}$
- DP limits information to reconstruct true answer from noisy answer
- Interpretation 1: We know it comes from Alice $\rightarrow$ DP protects content
- Interpretation 2: Anonymous collection $\rightarrow$ DP protects linkability if text contains signature

What are we protecting?

Text vs label protection

- Protect text d (Sensitive content like drug use) or protect label y (Sensitive rating)?
- If text reveals label (e.g., bad review text + 0 stars), protecting label might be useless
- Or protect text only? (Given 0-star label, we know experience was horrible)
- Usually dependent → Better protect entire data-point

Using local DP in NLP

Privatizing latent representation

Privatizing latent representation

- Classifier expects feature vector $x \in \mathbb{R}^j$ and label vector $y \in \mathbb{R}^k$
- Alice turns document d into x , label into one-hot y
- Privatize both with DP locally before releasing

Binary vectors: Bag of Words

Standard NLP task: Feature extraction / representation learning

- Baseline: *Bag of words* with vocabulary V
- Feature vector dimension $j = |V|$
- Options:
 - Raw counts: $x \in \{0, 1, 2, \dots\}^j$
 - Presence/absence: $x \in \{0, 1\}^j$
- Label y : Single bit ($y \in \{0, 1\}$) or one-hot ($y \in \{0, 1\}^k$)
- Privatize $x \rightarrow \hat{x}$ and $y \rightarrow \hat{y}$ using RR
- Split budget ε into ε_1 (features) and ε_2 (label)
- Or spend $\varepsilon_i = \varepsilon / (j + k)$ on each bit

Crucial problem: Dimensionality $j = |V|$

More features (larger V) $\rightarrow$ smaller ε_i per bit $\rightarrow$ More noise introduced to privatized vector

- Example 1: $\varepsilon = 2$, 1 bit $\rightarrow$ flip prob ≈ 0.1192 , looks OK
- Example 2: $\varepsilon = 2$, 100 bits $\rightarrow \varepsilon_i = 0.02 \rightarrow$ flip prob ≈ 0.4950
 - Result: Each bit almost completely random
 - Chances to train meaningful classifier look almost impossible
- Solution in literature: Drastically increase ε (e.g., $\varepsilon = 200$)
- Consequence: Effectively non-private

Problem with large ε

- Question: What's wrong with $\varepsilon = 200$?
- Sensitive question: 'Did you take illegal drugs?'
- Privatize with randomized response $\varepsilon = 200$
- Probability of revealing truth:

$$\frac{1}{1 + \exp(-200)} \approx 0.\underbrace{9999 \dots 9999}_{86 \text{ times}}891610$$

- Standard calculator shows 1.0
- No one willing to participate in such survey

Real-valued vectors: Embeddings

- Deep neural networks produce useful learned representations
- Turn text into high-dimensional embeddings vector
- Protect text with local DP:
 - Turn document into dense embedding (e.g., local transformer)
 - Apply DP mechanism for real-valued queries (Laplace or Gaussian)
 - Share privatized vector $\hat{x}$
- Guarantees: (ϵ, δ) -DP

Clipping and Sensitivity

- Real-valued queries must have bounded range
- Provide mechanism with global ℓ_1 or ℓ_2 sensitivity
- Achieved by clipping embeddings to unit ball ($C = 1$)
- Clip each dimension i : $\bar{x}_i = \min\{x_i, C\}$ or $\max\{x_i, -C\}$
- ℓ_1 sensitivity: $2C \cdot j$ (Manhattan distance)
- ℓ_2 sensitivity: $2C \cdot \sqrt{j}$
- Price for privacy: Higher dimension $j \rightarrow$ higher sensitivity $\rightarrow$ more noise

Using Privatized Embeddings

- Collect privatized embedding vectors from many users
- Vectors are in latent space ('a bunch of numbers')
- Use directly as feature vectors for training private classifier
- Provided we are also given (privatized!) labels
- Question: Can we provide something more useful than noisy text embedding?

Using local DP in NLP

Rewriting texts locally with DP
guarantees

Naive Text Rewriting Method

- Work on text level directly: 'privatize' locally to readable text
- Fix vocabulary V of $|V|$ words (include **NOWORD**)
- Define utility function $u : V \times V \mapsto \mathbb{R}$
- Measure similarity (e.g., cosine similarity of static word embeddings)
- Score between -1 and 1
- Pre-compute utility values (e.g., 1.25 billion for 50,000 words)

Exponential Mechanism Setup

- Determine sensitivity Δ for exponential mechanism
- Example: $\Delta = 2$ (max difference between $u(w, z) = -1$ and $u(w, w) = 1$)
- Fix number of tokens, for example $l = 20$ (huge limitation)
- All variables (V, u, Δ, l) publicly known

Generating word-by-word

- Query Alice: "What is your word at position i ?"
- Alice uses exponential mechanism (ε_i)
- Draw similar random word from V proportional to utilities
- Answer is $(\varepsilon_i, 0)$ -DP
- Repeat for l positions
- Overall budget $\varepsilon = \sum_{i=1}^l \varepsilon_i$

DP Guarantees and Reconstruction

- Algorithm is sound: l tokens privatized by local $(\epsilon, 0)$ -DP
- Probability of reconstructing true tokens D bounded by:

$$\frac{\Pr[D|\mathcal{M}(D) = y]}{\Pr[D'|\mathcal{M}(D') = y]} \leq \exp(\epsilon) \cdot \frac{\Pr[D]}{\Pr[D']}$$

- Takes into account priors $\Pr[D]$
- D' : Any other possible sequence of twenty tokens from vocabulary V
- Mental model: Extending randomized response (1 bit) to sequence of categorical values

Using local DP in NLP

Encoder-decoder models for private
rewriting

Encoder-decoder for Private Rewriting

- Original space of sequences is extremely high-dimensional
- Discrete DP algorithms (exponential mechanism) become infeasible
- Route: Work in lower-dimensional latent space (document embeddings)
- Unlike just privatizing embedding, use privatized latent vector to generate text

Transformer-based Approach

- Early transformers (T5, BART): Encoder + Decoder
- Entire input compressed into single real-valued vector
- Assumption: Latent vector can be clipped
- Privatize latent representation with DP (Laplace or Gaussian)
- Decoder rewrites back to text
- Advantage: Fluent text (vs. word-by-word random)
- Hope: Output similar to input (utility) and differentially private

DP-BART

- Example application: DP-BART
- Experimented with locally rewriting training data
- Tasks: Movie/product reviews, intent classification
- 'Pruning' latent vector to reduce sensitivity improved privacy/accuracy trade-off
- Result: Still ended up using absurdly large ϵ values
- Exceeded 200 for most datasets

Summary

Local DP: No trusted curator needed

Enables privatize raw data and publish with plausible deniability

Randomized response: Known and used since the 1960's

Relevance to NLP: Publishing data for training (but comes with a cost)

Privatizing text practically useless with pure DP, unless absurdly high ϵ , or something wrong, or very non-standard, or all of them

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